Support Vector Machines (SVM), SVR & SVM Kernels

Complete Notes with Detailed Step-by-Step Explanation

(Har chhoti se chhoti concept ko Hinglish mein, ekdum shuru se samjhaya gaya hai)

1. Introduction — Yeh Topic Kya Hai?

Machine Learning mein jab humein data ko do (ya zyada) groups mein baantna ho — jaise email **Spam hai ya Not Spam** — to isse **Classification** kehte hain. Aur jab humein ek number predict karna ho — jaise ghar ka **Price** uski **Size** se — to isse **Regression** kehte hain. In dono kaamo ke liye ek bahut powerful algorithm hai: **Support Vector Machine (SVM)**.

SVM ke do roop hote hain:

- **SVC (Support Vector Classifier)** → Classification ke liye
- **SVR (Support Vector Regressor)** → Regression ke liye

Is notes mein hum teeno cheezein samjhenge: (1) SVC ka concept aur uska maths, (2) SVR, aur (3) **SVM Kernels** — jo tab kaam aate hain jab data ek seedhi line se alag nahi ho paata.

2. Foundation — Logistic Regression Se SVM Tak

SVM samajhne se pehle ek chhota sa recap. **Logistic Regression** data ko ek **straight line** se do hisso mein baant deta hai (2D mein line, 3D mein plane, jahan axes x_1, x_2, x_3 hote hain). Notes ke pehle page mein upar wale orange × ek class hain aur neeche wale yellow × doosri class.

Problem yeh hai: aisi ek nahi, kai lines ho sakti hain jo dono classes ko alag kar deti hain. To *sabse achhi line kaun si hai?* SVM isi sawaal ka jawab deta hai — **wo line chuno jiske dono taraf ka gap (margin) sabse zyada ho.** Jitna bada gap, utna safe aur reliable classifier.

3. SVC — Marginal Planes, Support Vectors Aur Hard Margin

Notes ke hisaab se SVC ko step-by-step aise samjho:

Step 1 — Best Fit Line (Hyperplane)

Dono classes ke beech ek straight line kheenchi jaati hai. Isko **Hyperplane** ya **Decision Boundary** kehte hain. 2D mein ye line hoti hai, 3D mein plane.

Step 2 — Marginal Planes

Is line ke dono taraf ek-ek parallel dotted line kheenchte hain — inhe **Marginal Planes** kehte hain. Ye us jagah tak jaati hain jahan tak sabse paas wala data point hai.

Step 3 — Support Vectors

Jo data points seedhe marginal plane par baithe hote hain (notes mein yellow circle se dikhaye gaye hain) unhe **Support Vectors** kehte hain. Ye bahut important hain kyunki **boundary sirf inhi points se decide hoti hai** — baaki points hata bhi do to line nahi badlegi.

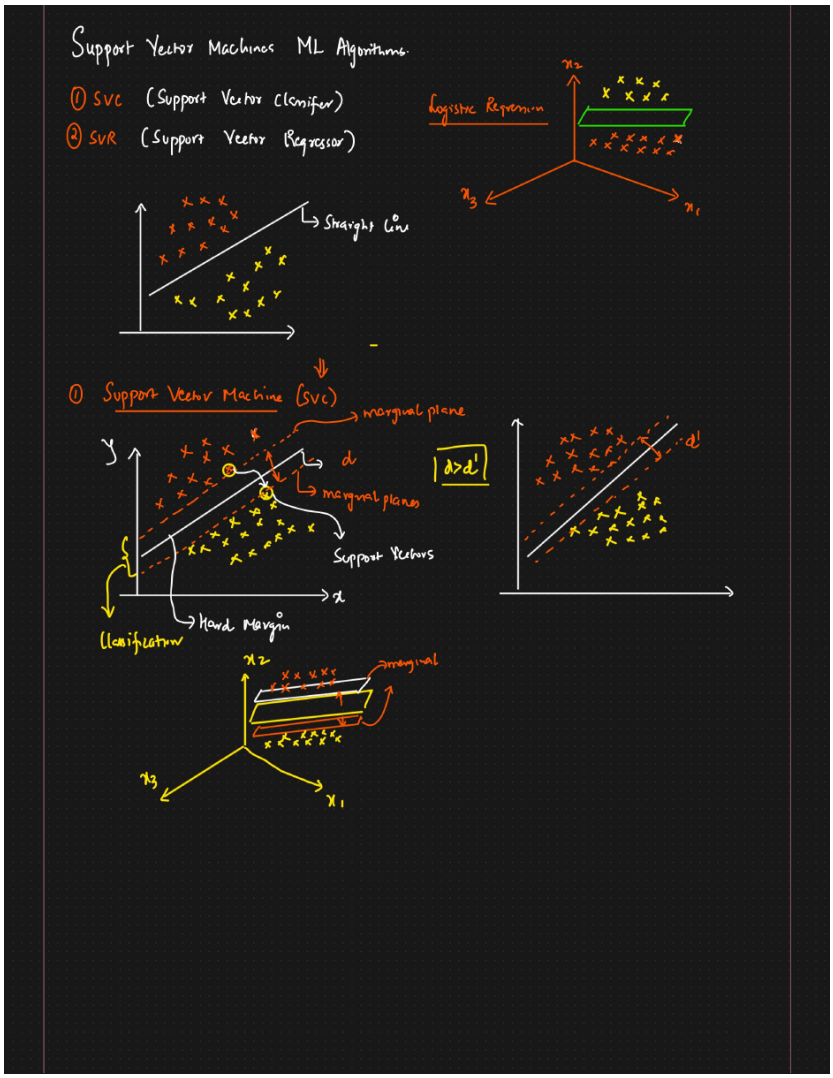
Step 4 — Margin Ka Size (d > d')

Line aur marginal plane ke beech ki doori ko **d** kehte hain. Notes mein do alag lines dikhayi gayi hain — ek mein gap bada (d), ek mein chhota (d'). Yahan condition likhi hai **d > d'**, matlab jis line ka gap bada hai wahi better hai, aur SVM wahi line chunta hai.

Step 5 — Hard Margin

Jab data bilkul clean ho aur dono classes ek straight line se perfectly alag ho sakti hon — koi bhi point marginal planes ke andar ya galat side na aaye — to isse **Hard Margin** kehte hain. Ye bohot strict hota hai, koi galti allowed nahi.

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Isme dikhaya gaya hai: SVM ke do types (SVC aur SVR), Logistic Regression ka straight line/plane, marginal planes, support vectors, d > d' ka concept, hard margin, aur 3D mein marginal planes.

4. Soft Margin vs Hard Margin

Real life ka data kabhi itna clean nahi hota. Notes ke doosre page mein dikhaya hai ki dono classes ke points thode **overlap** kar rahe hain (right side wala figure to bilkul mixed data hai). Aisi halat mein Hard Margin kaam hi nahi karta, kyunki ek seedhi line se perfect separation possible nahi.

Isliye **Soft Margin** use hota hai — isme hum **kuch points ko marginal planes ke andar ya galat side jaane ki ijazat dete hain**, taaki margin bada (maximum) rahe aur model overfit na ho. Matlab thodi galtiyan maaf, lekin overall boundary zyada general.

	Hard Margin	Soft Margin
Data	Perfectly separable, clean data	Overlapping / noisy data
Galti (misclassification)	Bilkul allowed nahi	Thodi allowed hai
Outliers ka effect	Bahut zyada (boundary hil jaati hai)	Kam
Cost function mein	Sirf $\|\|w\|\|/2$	$\|\|w\|\|/2 + C \sum \xi_i$

5. SVC Ka Maths Intuition

Ab dekhte hain ki ye margin nikalta kaise hai. Notes ka flow ye hai:

5.1 Line Ki Equation Vector Form Mein

School wali line ki equation hai **ax + by + c = 0**. Isse ML ke notation mein likhte hain:

$$w_1 x_1 + w_2 x_2 + b = 0 \quad \Rightarrow \quad w^T x + b = 0$$

Yahan **w** weights ka vector hai aur **b** bias/intercept. Agar line origin (0,0) se guzar rahi ho to **b = 0** ho jaata hai aur equation sirf $w^T x = 0$ reh jaati hai.

5.2 w Vector Plane Ke Perpendicular Hota Hai

Notes ke figure mein w ko plane se **90°** par dikhaya gaya hai. Isse **Normal Vector** kehte hain — ye plane ki direction batata hai. Kisi point ki plane se doori isi w par projection lekar nikalte hain.

5.3 Doori Ka Sign (+ve / −ve)

Notes ke example mein do points hain: **S(−4, 0)** aur **S'(4, 3)**.

- Jo point plane ke **neeche** hai → doori **d = −ve** (negative class)
- Jo point plane ke **upar** hai → doori **d = +ve** (positive class)

Matlab sirf sign dekh ke pata chal jaata hai ki point kis class mein hai: $w^T x + b > 0$ to positive side, < 0 to negative side.

5.4 Marginal Planes Ki Equations

Beech wali line (decision boundary) $\mathbf{w^T x + b = 0}$ hai. Uske upar wala marginal plane $\mathbf{w^T x + b = +1}$ aur neeche wala $\mathbf{w^T x + b = -1}$ maana jaata hai. (+1 aur −1 isliye chune gaye kyunki ye symmetrical hain aur calculation easy ho jaati hai.)

5.5 Margin Ka Formula Kaise Aaya (Step-by-Step)

Maan lo x_1 upar wale (+1) plane par hai aur x_2 neeche wale (−1) plane par:

$$\mathbf{w^T x_1 + b = +1}$$
$$\mathbf{w^T x_2 + b = -1}$$

Step 1: Dono equations ko subtract karo. Dekho, **b cancel ho jaata hai** (notes mein b par cross lagaya hai):

$$\mathbf{w^T (x_1 - x_2) = 2}$$

Step 2: Ab dono side ko **||w||** (w ki length/magnitude) se divide karo:

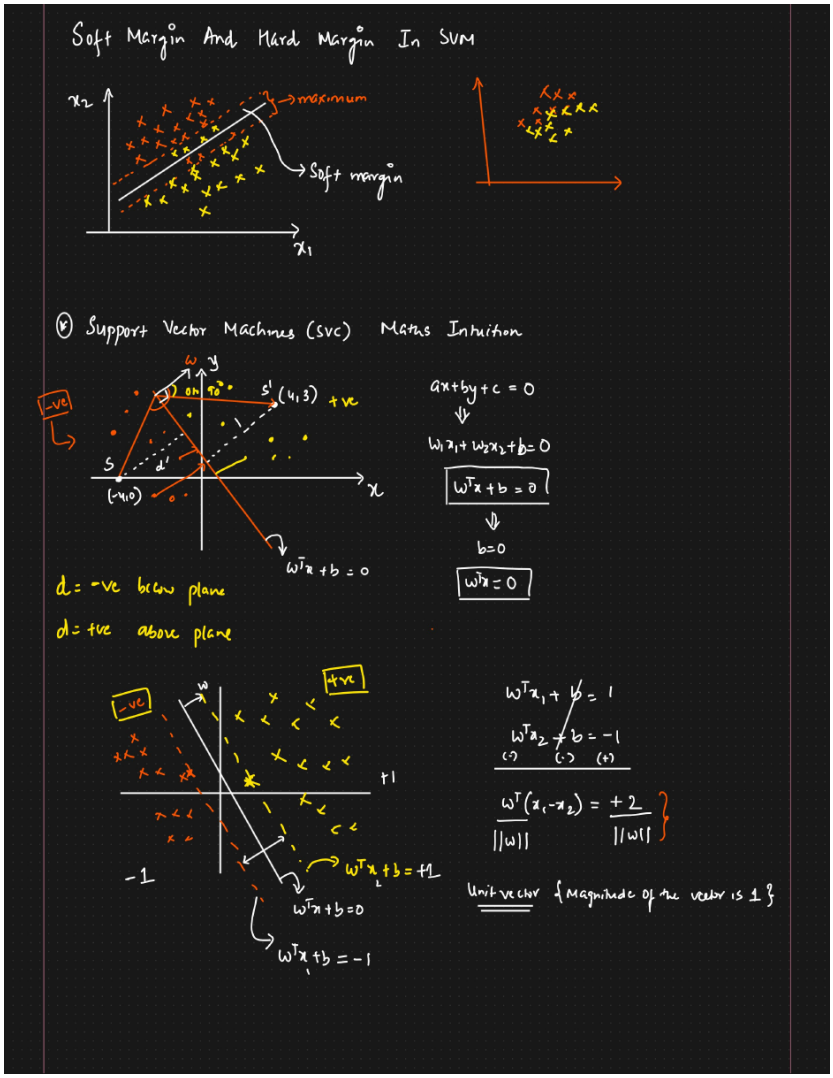
$$\mathbf{w^T (x_1 - x_2) / ||w|| \ = \ 2 / ||w||}$$

Ye kyun kiya? Kyunki **w / ||w||** ek **Unit Vector** hai (jiski magnitude 1 hoti hai — notes mein bhi yahi likha hai). Unit vector se dot product karne ka matlab hai us direction mein actual doori nikalna. Isliye left side ka matlab hai **dono marginal planes ke beech ki doori**, aur wahi hai:

$$\mathbf{Margin = 2 / ||w||}$$

Yaad rakho: **||w|| jitna chhota, margin utna bada.** Isi baat se cost function banta hai (agla section).

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Isme dikhaya gaya hai: Soft Margin vs Hard Margin ka figure, SVC ka Maths Intuition — line ki equation ($w^T x + b = 0$), w ka plane ke perpendicular hona, points S aur S' ka +ve/−ve distance, aur margin ke 2/||w|| tak pahunchne ka derivation.

6. SVC Ki Cost Function

6.1 Hard Margin Ka Objective

Humein **margin (2/||w||) ko maximum** karna hai, w aur b ki values change karke:

$$\textbf{Maximize}_{(w,\,b)} \quad 2 / ||w|| \quad \textbf{(= dono marginal planes ke beech ki doori)}$$

Constraint (shart): margin bada karne ke chakkar mein points galat side na chale jaayein. Correctly classified points ke liye:

- Agar $y_i = +1$ hai → $w^T x + b \geq 1$
- Agar $y_i = -1$ hai → $w^T x + b \leq -1$

6.2 Dono Conditions Ko Ek Mein Jodna

Dono conditions ko ek line mein likh sakte hain. Kyunki $y_i = +1$ ho to $(\geq 1) \times (+1) \geq 1$, aur $y_i = -1$ ho to $(\leq -1) \times (-1) \geq 1$. To **sabhi correct points ke liye** constraint:

$$y_i \times (w^T x_i + b) \geq 1$$

6.3 Maximize Se Minimize

2/||w|| ko maximize karna aur ||w||/2 ko **minimize** karna ek hi baat hai (kyunki ye ek-doosre ke reciprocal hain), aur minimize karna mathematically easy hota hai. Isliye:

$$\text{Maximize}_{(w,b)} \; 2/||w|| \;\Rightarrow\; \text{Min}_{(w,b)} \; ||w|| / 2$$

6.4 Soft Margin Ki Cost Function (Asli SVC)

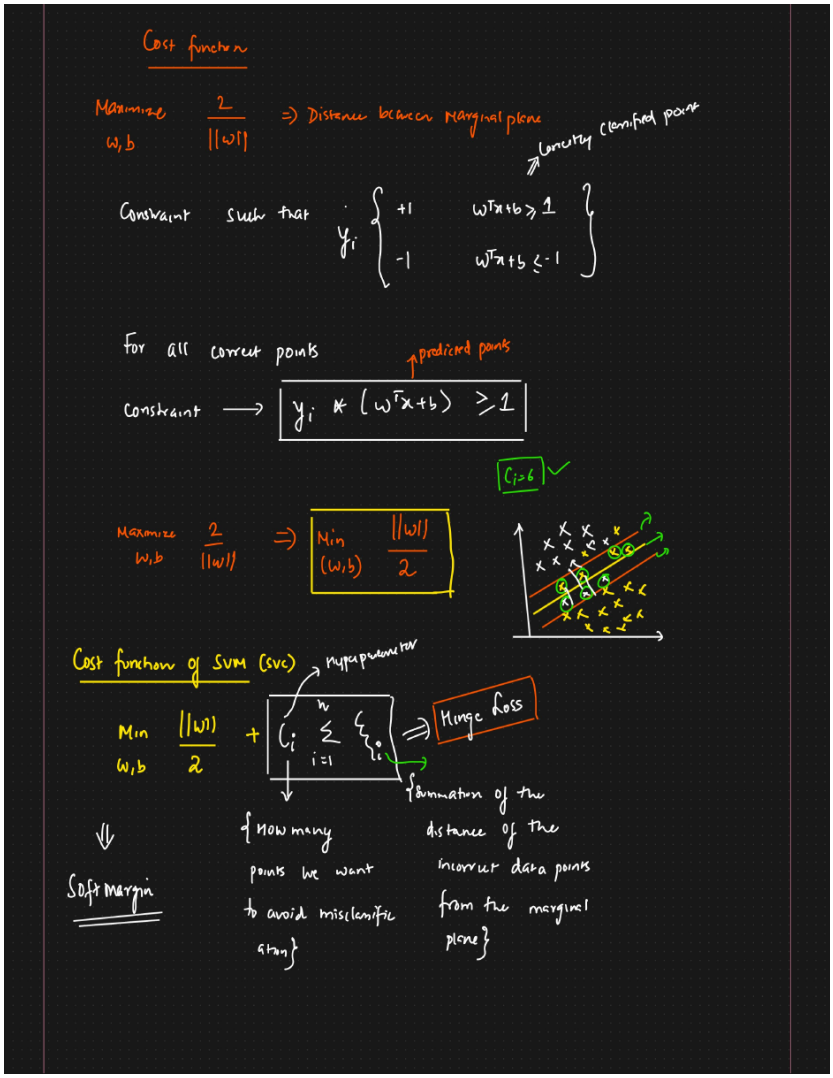
Real data mein kuch points galat jaate hi hain (notes ke figure mein green circle wale points marginal planes ke andar/galat side hain). Unhe handle karne ke liye cost function mein ek **penalty term** add karte hain:

$$\text{Min}_{(w,b)} \quad ||w|| / 2 \; + \; C \; \sum_{i=1}^{n} \xi_i$$

- **||w|| / 2** → margin ko bada rakhne wala part.
- **ξ_i (xi / zeta)** → galat classify hue points ki marginal plane se doori ka summation, yaani total error. Sahi points ke liye $\xi = 0$. Is poore $\sum\xi$ part ko **Hinge Loss** kehte hain.
- **C** → ek **hyperparameter** (hum khud set karte hain). Ye batata hai ki hum **kitne points ko misclassification se bachana chahte hain**.

C ka asar (extra samajh ke liye): C bada → model galti se bahut darta hai, margin chhota ho jaata hai aur overfitting ka khatra badhta hai. C chhota → kuch galtiyan allowed, margin bada, model zyada general. Ye cost function hi **Soft Margin** SVC kehlaati hai (notes mein neeche Soft Margin likha hai).

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Isme dikhaya gaya hai: Cost function ka derivation — Maximize 2/||w||, constraints $y_i(w^T x + b) \geq 1$, Maximize se Minimize ||w||/2, aur Soft Margin cost function (C aur ξ_i = Hinge Loss).

7. SVR — Support Vector Regression

SVC classes alag karta hai, jabki **SVR ek continuous number predict karta hai**. Notes ka example hai: **Size** (x-axis) se **Price** (y-axis) predict karna.

7.1 ε (Epsilon) Tube

Beech mein best-fit line **$w^T x + b$** hoti hai (orange). Uske upar aur neeche **ε** ki doori par do aur lines hoti hain (yellow):

- Upar wali line: **$w^T x + b + \varepsilon$**
- Neeche wali line: **$w^T x + b - \varepsilon$**

In dono lines ke beech ke area ko **ε-tube** ya **Marginal Error** zone kehte hain. Notes mein likha hai: ε = Marginal Error.

7.2 Tube Ke Andar Aur Bahar Ke Points

- **Tube ke andar** ke points → inka error ignore kiya jaata hai (koi penalty nahi). Prediction ε ke andar hai to chalega.
- **Tube ke bahar** ke points (notes mein yellow circle) → inpar penalty lagti hai. Tube se is point ki vertical doori ko $\boldsymbol{\xi_i}$ kehte hain (notes: ξ_i = error above the margin).

7.3 SVR Ki Cost Function Aur Constraint

$$\mathbf{Min_{(w,b)} \quad ||w|| / 2 \; + \; C \; \sum_{i=1}^{n} \xi_i}$$

Constraint (notes ke hisaab se):

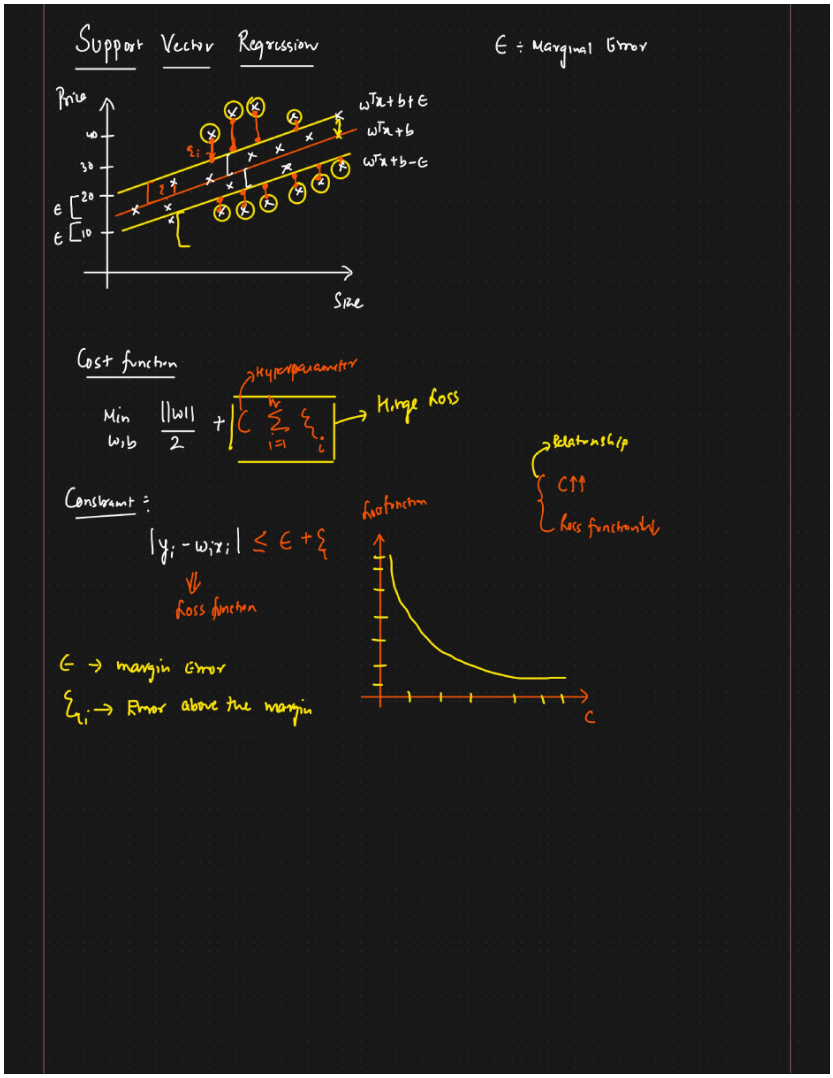
$$\mathbf{| \, y_i - w \, x_i \, | \leq \varepsilon + \xi_i}$$

Matlab: actual value (y_i) aur predicted value ka farak $\boldsymbol{\varepsilon + \xi_i}$ se zyada nahi hona chahiye. (Notes mein simple form likha hai; poori form mein predicted value $w^T x_i + b$ hoti hai.) Structure bilkul SVC jaisa hai — bas yahan ξ tube se bahar ke points ka error hai, aur ye **loss function** ka kaam karta hai.

7.4 C Aur Loss Ka Relation

Notes ke graph mein x-axis par **C** aur y-axis par **cost/loss function** hai. Curve neeche ki taraf aata hai, matlab **C ↑ ⇒ Loss function ↓**. Samajh: C bada karne par bahar ke points ki penalty ka weight badh jaata hai, to model unhe tube ke andar laane ki zyada koshish karta hai aur training loss kam hota hai (lekin bahut bada C overfitting la sakta hai).

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Isme dikhaya gaya hai: Support Vector Regression ka figure (Size vs Price, ε-tube, ξ_i), SVR ki Cost Function, constraint $|y_i - w\,x_i| \le \varepsilon + \xi$, aur C aur loss function ka graph.

8. SVM Kernels

8.1 Problem — Linear SVC Kab Fail Hota Hai?

Linear SVC sirf **straight line / plane** se data alag kar sakta hai. Notes ke pehle figure mein data ek line se alag ho jaata hai, to **Accuracy ↑** (binary classification). Lekin agar data aisa ho ki ek class andar ho aur doosri uske chaaron taraf (jaise circle ke andar aur bahar), to koi bhi straight line use alag nahi kar sakti — **Accuracy ↓**.

8.2 Solution — Kernel Transformation

Idea simple hai: agar 2D mein data alag nahi ho raha, to **use ek higher dimension mein bhej do**. Ek **mathematical formula** (transformation) lagake 2D data ko 3D mein convert kar dete hain. Notes ke figure mein dikhaya hai ki 3D mein andar wali (red) class upar uth jaati hai aur bahar wali (white) neeche, aur ab ek **plane (Linear SVC)** se aaram se alag ho jaati hain — **Accuracy ↑↑**. Isi transformation ka kaam **SVM Kernels** karte hain.

8.3 Simple Example — 1D Se 2D

Maan lo dataset **1D** hai: number line par points hain, aur ek class beech (0 ke paas) hai, doosri dono taraf. Ek single point (divider) se in dono ko alag nahi kar sakte. Ab transformation lagao:

$$y = x^2$$

Ab har point ka ek naya coordinate (x, x^2) ban gaya, yaani data **2D** mein aa gaya. Beech wale points neeche rahe (x^2 chhota) aur bahar wale points upar chale gaye (x^2 bada), jisse ab ek seedhi line (Linear SVC) dono ko alag kar deti hai — **Accuracy ↑**.

8.4 Kernels Ke Types

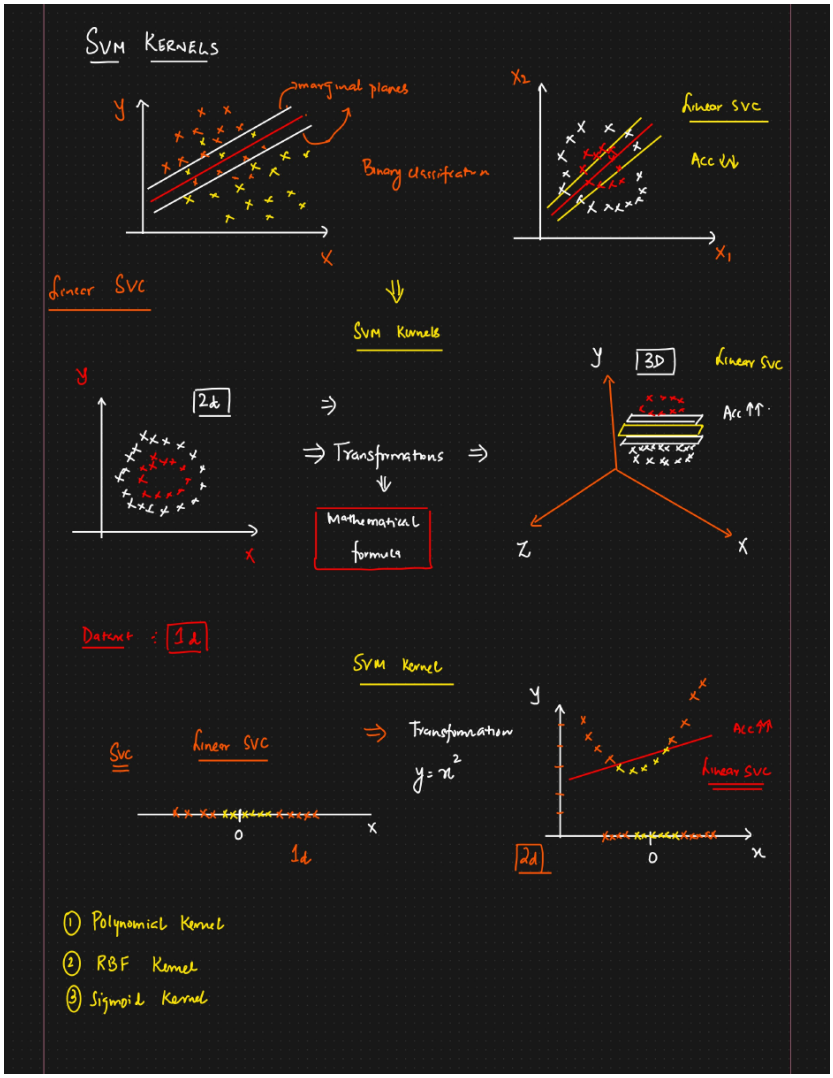
Notes mein teen main kernels likhe hain:

Kernel	Formula (extra info)	Kab / Kaise kaam karta hai
1. Polynomial Kernel	$K(x, z) = (\gamma\, x^T z + r)^d$	Data ko polynomial (curved) features mein transform karta hai; d = degree.
2. RBF Kernel	$K(x, z) = \exp(-\gamma\, \|\|x - z\|\|^2)$	Sabse popular. Do points ki doori ke basis par similarity naapta hai; complex, round-type boundaries bana sakta hai.
3. Sigmoid Kernel	$K(x, z) = \tanh(\gamma\, x^T z + r)$	Neural network wale tanh/sigmoid function jaisa behave karta hai.

Note: Teacher ke notes mein sirf in teen kernels ke naam hain; formulas aur short explanation maine revision ke liye extra jode hain.

Kernel Trick: asal mein SVM data ko sach mein higher dimension mein convert nahi karta; kernel function seedha dot product nikal deta hai jaise data higher dimension mein ho. Isse computation bhi fast rehta hai.

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Isme dikhaya gaya hai: Linear SVC ka 2D data par fail hona, SVM Kernels ke through 2D se 3D transformation, 1D dataset ko $y = x^2$ se 2D mein le jaana, aur teen kernels ke naam (Polynomial, RBF, Sigmoid).

9. Quick Summary Table (Revision Ke Liye)

	SVC	SVR
Full form	Support Vector Classifier	Support Vector Regressor
Kaam	Classification (class predict)	Regression (number predict)
Margin	Marginal planes: $w^T x + b = \pm 1$	ε-tube: $w^T x + b \pm \varepsilon$
Cost function	$\|\|w\|\|/2 + C \sum \xi_i$	$\|\|w\|\|/2 + C \sum \xi_i$
ξ_i kya hai?	Galat classify points ki marginal plane se doori	Tube ke bahar ke points ka error
Hyperparameters	C (aur kernel)	C, ε (aur kernel)

	Hard Margin	Soft Margin	Kernels
Idea	Zero galti, strict	Thodi galti allowed (C se control)	Non-linear data ko higher dimension mein bhejna
Kab use	Perfectly separable data	Overlapping / real data	Jab straight line se data alag na ho
Examples	—	—	Polynomial, RBF, Sigmoid

Ek line mein yaad rakhna: SVM wo line/plane dhoondta hai jiska margin **2/||w||** sabse bada ho; **C** batata hai kitni galtiyan allowed hain (soft margin), **ε** SVR mein error-tube ki chaudai hai, aur agar data seedhi line se alag nahi hota to **Kernel** (Polynomial / RBF / Sigmoid) use karke data ko higher dimension mein bhej do.